SRI LANKA DISASTER-MANAGEMENT GAME

Recommended Game Structure and Evidence-Validated Revised Questions

Flood + drought compound-disaster decision simulation

Purpose

This document converts the original six-bucket question set into a defensible game scenario sequence. Each player-facing premise and action is tagged as either (1) directly supported by historical/official evidence, (2) a realistic transformation of that evidence for gameplay, or (3) a fictional game constraint that must not be presented as a historical fact.

Prepared for implementation in the existing MongoDB Action / Scenario / Event configuration

Validation date: 25 September 2026

1. Executive Summary

Recommended progression: the game should move from information gathering, to compound-hazard interpretation, to resource allocation, to logistics, to life-safety decisions, and finally to recovery and needs assessment. This creates a coherent decision-making arc rather than six unrelated questions.

Key validation rule

The game may use fictional budgets, truck counts, probabilities, impact points and dates. What must be accurate is the underlying disaster logic: Sri Lanka has experienced severe flood/landslide events, prolonged drought and water stress, simultaneous disruption across transport/communications/WASH/health, and formal post-disaster needs assessment processes. [R1-R11]

Stage	Time	Decision Theme	Player Skill	Primary Scoring Anchor	Secondary Defensible Choice
1	Day 0	Early warning + uncertainty	Information gathering before irreversible spending	1A	1D
2	Day 2	Compound flood + drought	Fuse official data and verified local observations	2B	2E
3	Day 4	Budget constraint	Allocate scarce funds explicitly and show the trade-off	3A	3D
4	Day 6	Transport constraint	Use finite logistics capacity without unsafe overload	4A	4E
5	Day 8	Life safety + access	Protect people in flood/landslide zones while maintaining a multi-hazard response	5A	5C
6	Day 10	Recovery transition	Assess damage/needs, restore WASH/transport/health and mobilize recovery	6A	6C/6E

2. Evidence and Validation Standard

The question set uses three evidence levels:

- Level A - Direct historical/official fact: the statement is explicitly supported by an official or intergovernmental source (for example, the 2017 PDNA, DMC flood records, or the 2025 Cyclone Ditwah PDNA).
- Level B - Realistic game translation: the scenario changes location, timing, quantities or sequencing but preserves a documented disaster mechanism (for example, contaminated water after flooding or isolated communities after road disruption).
- Level C - Fictional game parameter: budget amounts, truck inventory, exact player choices and impact scores are constructed for gameplay and are not historical claims.

Important: The recommended scoring anchors are game-design choices under the stated fictional constraints. They are not universal instructions for real disaster commanders; real response decisions depend on current warnings, populations at risk, operational capacity, access, and professional emergency-management coordination.

3.1 S1_START - Day 0: Early Warning and Uncertainty

Decision theme: Information before irreversible spending

Player-facing question

QUESTION

A developing low-pressure system over the Bay of Bengal may produce heavy rainfall over parts of Sri Lanka during the coming days. Forecast confidence is moderate. At the same time, some dry-zone communities report continuing water stress after a prolonged dry period. You have a fictional emergency allocation of LKR 500 million. What should you do first?

Validation of the scenario facts

Level A: Sri Lanka has a distinct October-November Second Inter-Monsoon, during which depressional activity can contribute substantially to rainfall. November is therefore a plausible period for a game scenario involving a low-pressure/depression signal. [R10]

Level A: The 2016 drought continued for almost two years and motivated rural water-supply interventions in the Dry Zone, including Anuradhapura and Polonnaruwa. [R3]

Level B: Combining an uncertain rainfall threat with pre-existing dry-zone water stress is a realistic compound-risk setup inspired by Sri Lanka's experience of alternating and overlapping hazards. [R3, R4]

Level C: LKR 500 million is a game parameter and is not presented as a historical emergency-budget figure.

Revised actions and fact basis

ID	Player action	Why this action is realistic / what fact it is based on	Game role
1A	Inspect water sources and identify vulnerable communities	A preliminary information/needs check is a low-regret action. Sri Lanka's disaster planning and recovery processes emphasize data collection, local assessment and needs analysis. [R1, R4]	Primary scoring anchor
1B	Purchase a large emergency stock of bottled water immediately	Emergency water is legitimate, but the broad purchase is intentionally expensive before the hazard magnitude is clear. Water access is a documented concern in Dry Zone communities. [R3]	Useful but over-commits budget
1C	Clear priority drainage channels in flood-prone communities	Drainage maintenance is directly aligned with flood-risk management. The Irrigation Department identifies drainage/flood-system maintenance and rehabilitation as core flood-management functions. [R6]	Defensible preventive action

ID	Player action	Why this action is realistic / what fact it is based on	Game role
1D	Communicate uncertainty and preparedness guidance to communities	Sri Lanka has used early warning and public communication through media, police, district disaster networks and other channels. [R11]	Secondary scoring anchor
1E	Wait for a more certain forecast before acting	Delaying all action sacrifices lead time; official flood-warning practice is explicitly designed to provide advance notice for preparedness. [R6, R11]	Intentionally weak choice

Recommended scoring interpretation

SCORING

Recommended anchor: 1A. Secondary defensible choice: 1D or 1C depending on the player's stated priority. Do not score 1A as the only real-world correct answer; use the game's stated objective of uncertainty reduction.

3.2 S2_DUAL - Day 2: The Compound Hazard

Decision theme: Flood response and dry-zone water stress at the same time

Player-facing question

QUESTION

Rainfall in one part of the country has reached approximately 150 mm in 24 hours, creating localized flooding. At the same time, dry-zone communities report continuing water shortages. Which information and response strategy should you adopt?

Validation of the scenario facts

Level A: Cyclone Ditwah produced rainfall of 150-500 mm across affected areas and exceeded 300 mm within 24 hours in several districts, with widespread flooding and landslides. [R5]

Level A: Sri Lanka's Dry Zone has experienced prolonged drought and water-security challenges, and Anuradhapura and Polonnaruwa are specifically identified within dry-zone water-management initiatives. [R3, R4]

Level B: The exact combination and timing of 150 mm/24h rainfall plus dry-zone shortage is a synthesized game event, not a claim that these exact two conditions always occurred simultaneously in one historical event.

Revised actions and fact basis

ID	Player action	Why this action is realistic / what fact it is based on	Game role
2A	Depend mainly on social-media reports because they arrive fastest	Unverified information is a poor basis for resource allocation. Disaster response depends on official warnings, verified field reports and monitoring. [R6, R11]	Weak information strategy
2B	Combine official forecasts, river/reservoir observations	This directly matches the multi-source monitoring model used	Primary scoring anchor

ID	Player action	Why this action is realistic / what fact it is based on	Game role
	and verified local reports	in Sri Lankan hazard management. The DMC and Irrigation Department provide weather, river-level and warning information. [R6, R11]	
2C	Spend most of the emergency budget immediately on both hazards	Fast spending may be justified in a confirmed emergency, but the action is intentionally framed as a large commitment before all needs are measured.	Overreactive under uncertainty
2D	Wait until flooding becomes severe before activating response measures	This loses warning lead time. Sri Lanka's flood-warning system is designed to enable advance action. [R6, R11]	Weak
2E	Establish a district-level monitoring network involving local authorities and community observers	Community and local coordination is consistent with participatory assessment and disaster-network practice. [R1, R11]	Secondary scoring anchor

Recommended scoring interpretation

SCORING

Recommended anchor: 2B. Secondary defensible choice: 2E. A strong implementation can give partial credit when a player selects both through a multi-selection mechanic, but keep 2B as the single-choice anchor.

3.3 S3_CALC - Day 4: The LKR 100 Million Constraint

Decision theme: Explicit trade-off allocation

Player-facing question

QUESTION

Three immediate requirements are identified: LKR 40M for Kelani-area evacuation support, LKR 50M for dry-zone drinking-water response, and LKR 30M for emergency medical supplies. Total identified requirements are LKR 120M, but the fictional allocation available this round is only LKR 100M. You cannot fully fund all three. How do you allocate the available funds?

Validation of the scenario facts

Level A: Historical Sri Lankan disaster assessments show that floods and landslides can create concurrent needs across evacuation/rescue, infrastructure, water, health and livelihoods. The 2017 PDNA covered nine sectors and included government, environment, disaster-risk reduction and other cross-cutting themes. [R1]

Level A: The 2017 recovery-planning process explicitly coordinated multiple ministries and sectors, including health, agriculture, water, irrigation, transport and disaster management. [R2, R7]

Level C: The LKR 40M/50M/30M requirements and LKR 100M ceiling are fictional arithmetic constraints intentionally created to test prioritization. They are not historical allocations.

Revised actions and fact basis

ID	Player action	Why this action is realistic / what fact it is based on	Game role
3A	Allocate LKR 40M evacuation + LKR 30M water + LKR 30M medical	Fully protects the stated evacuation and medical requirements while reducing the water request by LKR 20M. It demonstrates explicit prioritization across life safety, WASH and health.	Primary scoring anchor
3B	Allocate LKR 60M evacuation + LKR 40M water + LKR 0M medical	Maximizes two sectors but leaves no emergency medical allocation. This is a deliberate high-risk trade-off.	Weak/imbalanced
3C	Allocate LKR 40M evacuation + LKR 50M water + LKR 10M medical	Fully funds evacuation and water but leaves a medical shortfall.	Defensible alternative if health needs are lower than stated
3D	Allocate LKR 30M evacuation + LKR 40M water + LKR 30M medical	Maintains medical funding while underfunding both evacuation and water.	Secondary defensible choice if access/medical risk is emphasized
3E	Release the full LKR 120M despite having only LKR 100M	Violates the game's stated resource constraint and should be scored as infeasible.	Invalid under game rules

Recommended scoring interpretation

SCORING

Recommended anchor: 3A because it explicitly funds the stated life-safety and health minimums and exposes the 20M water trade-off. 3D can receive substantial partial credit; 3C can also receive partial credit. Do not claim that historical Sri Lanka had a single universally correct 40/30/30 allocation.

3.4 S4_LOGISTICS - Day 6: Fifty Trucks, Two Emergencies

Decision theme: Finite logistics capacity

Player-facing question

QUESTION

The Kelani basin is experiencing serious flooding while dry-zone communities require emergency drinking water. You have exactly 50 heavy-duty trucks nationwide. The flood-rescue plan needs 30 trucks and the water-delivery plan needs 20 trucks. How should the trucks be deployed?

Validation of the scenario facts

Level A: The Kelani River is a long-established flood-risk basin. The Irrigation Department has monitored the Nagalagam Street gauge since 1924, and historical records document repeated flood levels, including major and dangerous events. [R6, R7]

Level A: During major Sri Lankan flood events, rescue teams, boats, military forces and other logistics were mobilized, while road inundation could limit land-based access. [R1, R11]

Level B: Using exactly 30 + 20 = 50 trucks is a fictional capacity constraint that reproduces the real decision problem of allocating scarce transport across competing humanitarian needs.

Revised actions and fact basis

ID	Player action	Why this action is realistic / what fact it is based on	Game role
4A	Deploy 30 trucks for flood rescue and 20 for water delivery	Exactly satisfies both stated operational requirements without exceeding capacity.	Primary scoring anchor
4B	Deploy all 50 trucks for flood rescue	Maximizes flood response but abandons the simultaneous water requirement.	Hazard-specialist but imbalanced
4C	Deploy all 50 trucks for water delivery	Protects water supply but abandons immediate rescue capacity.	Hazard-specialist but imbalanced
4D	Attempt to deploy 60 trucks by using vehicles that have not completed maintenance	Introduces an explicitly unsafe and infeasible over-capacity option.	Invalid/unsafe
4E	Hold the trucks and wait for external transport assistance	External support can be useful, but delaying available local capacity risks losing lead time. [R1, R11]	Weak under immediate threat

Recommended scoring interpretation

SCORING

Recommended anchor: 4A. The strongest game mechanic is to make this a hard capacity check: allocations above 50 are infeasible, while allocations below 50 leave unmet demand that affects later state variables.

3.5 S5_DISASTER - Day 8: The Breaking Point

Decision theme: Life safety under a compound disaster

Player-facing question

QUESTION

Heavy rain has triggered landslides in parts of Ratnapura and Kalutara. At the same time, several dry-zone communities have severe drinking-water shortages, roads are difficult to access, and parts of the telecommunications network have failed. What should be prioritized?

Validation of the scenario facts

Level A: Ratnapura and Kalutara were among the worst-affected districts in the May 2017 floods and landslides; official recovery work included support for these districts. [R1, R12]

Level A: Cyclone Ditwah in 2025 caused widespread flooding and landslides, access constraints, infrastructure damage, water-supply disruption and communication difficulties. [R5, R8, R9]

Level B: The exact combination of Ratnapura/Kalutara + dry-zone water shortage + telecom failure in one Day-8 scenario is a synthesized compound-disaster exercise inspired by documented hazard interactions.

Revised actions and fact basis

ID	Player action	Why this action is realistic / what fact it is based on	Game role
5A	Order evacuation of people in identified high-risk flood and landslide zones	Life safety and evacuation are directly supported by Sri Lanka's disaster-response practice and past use of safe locations when warnings indicate danger. [R11]	Primary scoring anchor
5B	Use aircraft to airdrop drinking water to every affected dry-zone community	Aerial delivery can be useful for isolated locations, but applying it universally is inefficient. The game should reserve this option for inaccessible communities.	Conditional / situational
5C	Establish emergency coordination and triage points serving both hazard zones	Multi-sector coordination and rapid response are strongly supported by the 2017 and 2025 response models. [R1, R8]	Secondary scoring anchor
5D	Evacuate only residents who volunteer	Voluntary self-evacuation can leave people exposed where authorities identify high life-safety risk. The action is intentionally weak.	Weak under identified high risk
5E	Suspend non-essential government expenditure and redirect available funds to emergency response	Could release fiscal space but may create service disruption. Treat it as a strategic trade-off rather than an obviously bad action.	Secondary/conditional

Recommended scoring interpretation

SCORING

Recommended anchor: 5A for immediate life safety; 5C should receive substantial secondary credit because compound disasters require coordination. Do not imply that every real evacuation should be mandatory; the game assumption is that authorities have identified high-risk zones requiring protective action.

3.6 S6_AFTERMATH - Day 10: From Emergency Response to Recovery

Decision theme: Needs assessment, WASH, health and critical infrastructure

Player-facing question

QUESTION

The immediate weather threat is weakening. Initial assessments show damaged roads and bridges, contaminated or unsafe domestic water sources, damaged water-supply systems, disrupted communications, displaced households, and increased public-health risk. What should the administration do next?

Validation of the scenario facts

Level A: After Cyclone Ditwah, official and humanitarian assessments documented severe damage to water systems, transport/access, health facilities and other infrastructure. [R5, R8, R9]

Level A: The 2017 recovery process used a formal Post-Disaster Needs Assessment and a subsequent recovery plan covering multiple sectors and three cross-cutting themes. [R1, R2]

Level A: WHO's response to Ditwah included rapid response teams, disease surveillance and coordination across health, WASH and nutrition; this supports making health and WASH a core recovery decision. [R8]

Level B: The exact Day-10 damage list is a game synthesis. It is credible because the listed sectors were affected in documented events.

Revised actions and fact basis

ID	Player action	Why this action is realistic / what fact it is based on	Game role
6A	Conduct a rapid multi-sector damage and needs assessment, then prioritize WASH, health, transport and shelter recovery	This directly mirrors PDNA/JRNA practice: assess first, prioritize, then mobilize resources and recovery actions. [R1, R2, R5, R8]	Primary scoring anchor
6B	Restore critical roads, clear debris and rehabilitate damaged water infrastructure	Transport and water restoration are high-value recovery actions, especially where access constrains remote facilities. [R5, R9]	Strong operational follow-up
6C	Strengthen disease surveillance, safe water, sanitation and emergency health services	Strongly supported by WHO and UNICEF response evidence following Ditwah and by flood-related disease experience in 2017. [R8, R9, R10]	Strong secondary action
6D	Strictly ration all remaining food and water without reassessing community needs	Can be necessary in a true shortage, but doing so without updated needs data creates	Weak unless triggered by scarcity state

ID	Player action	Why this action is realistic / what fact it is based on	Game role
		avoidable inequity and operational risk.	
6E	Request external assistance and coordinate donor/partner support against a documented needs list	International and inter-agency support was part of Sri Lanka's response/recovery arrangements. The needs list is the important condition. [R1, R2, R8]	Strong secondary action

Recommended scoring interpretation

SCORING

Recommended anchor: 6A. 6B, 6C and 6E should produce meaningful positive effects and can be made synergistic in the game. This makes the final stage about recovery planning rather than a single “magic” action.

4. Consolidated Answer / Scoring Anchor Matrix

Use this table as the configuration reference for the game. “Primary anchor” means the intended single-choice high score under the exact fictional assumptions shown in the question; it does not mean that real disaster management has one universally correct answer.

Scenario	Primary anchor	Secondary choices	Core skill	Historical foundation	Game-only elements
S1_START	1A	1D, 1C	Information under uncertainty	Second inter-monsoon risk; drought/water stress; warning practice	LKR 500M
S2_DUAL	2B	2E	Multi-source risk interpretation	150+ mm rainfall is plausible and documented in Ditwah; Dry Zone water stress is documented	Exact simultaneous timing
S3_CALC	3A	3D, 3C	Budget trade-off	Multi-sector needs and recovery planning	All LKR values
S4_LOGISTICS	4A	4E (partial)	Finite logistics allocation	Kelani flood risk; rescue/transport constraints	Exactly 50 trucks and 30/20 split
S5_DISASTER	5A	5C, 5E	Life safety + coordination	Ratnapura/Kalutara 2017; compound flood/landslide/communications/WASH issues in 2025	Exact combined Day-8 configuration
S6_AFTERMATH	6A	6B, 6C, 6E	Recovery transition	PDNA/JRNA, WASH, health, transport recovery	Day-10 timing and impact amounts

5. Configuration Recommendations for the Existing MongoDB Data

Budget units

Use one unit consistently. Recommended: emergencyBudget is stored in LKR millions. Example: -40 means LKR 40M. Avoid mixing “500M” in narrative with “-10000” in impact data.

Recommended state variables:

- emergencyBudget
- floodPreparedness
- droughtPreparedness
- waterSecurity
- healthRisk
- foodSecurity
- transportation

- communications
- communityTrust
- informationQuality
- decisionQuality

Recommended event design:

- Keep KELANI_SPILL as a basin-specific flood event, but trigger it from river level / hazard state rather than from a generic bucket alone when possible. The Nagalagam Street gauge has real historical significance. [R6, R7]
- Replace MAHA_FAILURE ("total failure confirmed") with a crop-damage / food-security disruption event. Harvest failure is a season-level outcome and should not be "confirmed" immediately after the start of a November disaster.
- Add event effects for waterSecurity, healthRisk, foodSecurity and communications so the later questions depend on earlier actions rather than only changing a generic flood/drought score.
- Allow multiple actions to be defensible in some scenarios. For example, S5 should reward both immediate evacuation and strong multi-hazard coordination, while S6 should reward assessment plus operational recovery.

6. Final Realism Checklist Before Player Release

Check	Requirement
Historical claims	Every statement that sounds like a real Sri Lankan statistic must have a source or be clearly marked as hypothetical.
Dates	Use November 2026 as the fictional game setting, but describe it as a hypothetical scenario inspired by previous Sri Lankan events.
Weather terminology	November is the Second Inter-Monsoon, not the Northeast Monsoon. Depressions and heavy rainfall can still occur in this period. [R10]
Numbers	LKR budgets, truck counts, impact points and exact action costs are game mechanics unless a source explicitly supports them.
Locations	Ratnapura, Kalutara and the Kelani basin are strong historical anchors for flood/landslide/flood-warning gameplay. [R1, R6, R7]
Dry Zone	Anuradhapura and Polonnaruwa are valid dry-zone water/agriculture references, but exact community water shortages should be stated as scenario facts, not historical claims. [R3, R4]
Health	Flood recovery should include WASH and disease surveillance, because these were documented response priorities after Ditwah and during major flood events. [R8, R9]
Recovery	The final stage should use assessment -> prioritization -> recovery sequencing rather than simply "spend more money." [R1, R2, R5]
Scoring	Present "correct" answers as intended game scoring under the stated rules, not as a claim that Sri Lankan authorities would always choose exactly one action.

7. Evidence Sources

R1 - United Nations Development Programme (UNDP) Sri Lanka. Sri Lanka Rapid Post Disaster Needs Assessment 2017. 21 Nov 2017.

https://www.undp.org/srilanka/publications/sri-lanka-rapid-post-disaster-needs-assessment-2017

R2 - UNDP Sri Lanka. Sri Lanka Post Disaster Needs Assessment 2016.

https://www.undp.org/srilanka/publications/sri-lanka-post-disaster-needs-assessment-2016

R3 - UNDP Sri Lanka. Rural Water Supply Schemes for Climate-Vulnerable Communities in the Dry-Zone. 7 Apr 2021.

https://www.undp.org/srilanka/press-releases/rural-water-supply-schemes-climate-vulnerable-communities-dry-zone

R4 - FAO/WFP. Special Report - Crop and Food Security Assessment Mission to Sri Lanka. 22 Jun 2017.

https://www.fao.org/in-action/kore/publications/publications-details/en/c/899951/

R5 - Government of Sri Lanka, Disaster Management Centre. Cyclone Ditwah, Sri Lanka 2025 - Post-Disaster Needs Assessment, Full Report. Published/updated 2026.

https://www.dmc.gov.lk/images/pdfs/PDNA_Full_Report_Final%20Updated.pdf

R6 - Irrigation Department Sri Lanka. Hydrology & Disaster Management - Introduction / hydrometeorological information and flood warning functions.

https://www.irrigation.gov.lk/web/index.php?Itemid=&id=27&lang=en&option=com_content&view=article

R7 - Irrigation Department Sri Lanka. Hydrological Annual 2017/18 - Kelani Ganga flood history and Nagalagam Street gauge.

https://www.irrigation.gov.lk/web/images/Hydrological-Annual/11_Hydrological_2017-2018.pdf

R8 - World Health Organization (WHO) Sri Lanka. Cyclone Ditwah 2025 - emergency response and health risks.

https://www.who.int/srilanka/emergencies/cyclone-ditwah-2025

R9 - UNICEF Sri Lanka. Humanitarian Situation Report No. 5 - Cyclone Ditwah, 26 Dec 2025.

https://www.unicef.org/media/177696/file/Sri-Lanka-Humanitarian-Situation-Report-No.5-%28Cyclone-DITWAH%29%2C-26-December-2025.pdf

R10 - Department of Census and Statistics Sri Lanka. Statistical Abstract - climate/rainfall seasons.

https://www.statistics.gov.lk/abstract2024/chapter1

R11 - Disaster Management Centre Sri Lanka. Annual Report 2017 - warnings, evacuation, rescue and landslide response.

https://www.dmc.gov.lk/images/pdfs/AnnualReports/2017_compressed.pdf

R12 - UNDP Sri Lanka. Building Back Better - 2017 floods and landslides, affected districts and resilience programmes.

https://www.undp.org/srilanka/stories/building-back-better

R13 - Disaster Management Centre Sri Lanka. 28 Nov 2025 river water levels and rainfall situation report.

https://www.dmc.gov.lk/images/dmcreports/Situation_Summary_Report_at_1800hrs_on_2025__1764337168.pdf

8. Release Recommendation

Recommended release framing

“This game is a hypothetical Sri Lankan disaster-management simulation inspired by documented floods, landslides, droughts, river flooding and Cyclone Ditwah. Historical facts are used to shape the hazard mechanisms; budgets, inventories, exact timings and score values are fictional game parameters.”